

Modeling of Temperature Dependent Composite Buckling with FEM

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Abstract

This study investigates the thermal buckling behavior of carbon/epoxy composite laminates using eigenvalue and nonlinear finite element analyses in ANSYS with temperature-dependent material properties derived from micromechanical models, including the Rule of Mixtures, Chamis relations, and Schapery's thermal expansion formulation. A parametric study of symmetric laminate stacking sequences with 0° , 90° , and $\pm 45^\circ$ plies across varying plate geometries showed that stacking sequence, plate dimensions, and outer ply orientation strongly influence thermal buckling resistance. Eigenvalue analyses predicted higher critical buckling temperatures due to linear assumptions, while nonlinear analyses incorporating geometric imperfections and material degradation produced substantially lower critical temperatures and, in some cases, no buckling within the investigated temperature range, indicating that material softening may dominate the response and lead to alternative failure modes. Overall, the results demonstrate the limitations of eigenvalue buckling analysis for thermally loaded composites and highlight the importance of nonlinear analysis for more realistic predictions and laminate optimization.

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1 Keywords

Keywords: Composite laminates; Thermal buckling; Finite element analysis; Eigenvalue analysis; Nonlinear analysis; Temperature-dependent properties; Micromechanics; Stacking sequence optimization

2 Introduction

Carbon fiber reinforced epoxy composites are widely used in aerospace structures due to their high specific stiffness, strength, and low weight. These materials are commonly employed in applications subjected to elevated thermal environments, including hypersonic vehicles, propulsion systems, and directed energy weapon platforms. Under severe thermal loading, constrained thermal expansion can generate compressive stresses which may lead to thermal buckling, a critical structural instability failure mode.

Predicting the thermal buckling response of composite laminates is challenging due to the anisotropic nature of composite materials and the temperature dependence of matrix-dominated properties. In particular, matrix softening at elevated temperatures can significantly reduce laminate stiffness and alter buckling behavior. However, many existing studies rely on linear eigenvalue buckling analyses and temperature-independent material properties, which may lead to overly high critical buckling temperatures.

The present study investigates the thermal buckling behavior of carbon/epoxy composite laminates using both eigenvalue and nonlinear finite element analyses in ANSYS. Temperature-dependent composite properties were determined using micromechanical models, and the effects of laminate stacking sequence and plate geometry on thermal buckling response were evaluated. The results provide insight into the limitations of linear buckling analysis and the influence of material degradation on composite thermal stability.

3 Literature Analysis

Wei Zhao et al. [1] investigated the thermal buckling behavior and optimization of curvilinearly stiffened composite plates using variable angle tow (VAT) laminates. Their work focused on improving buckling performance through the use of spatially varying fiber orientations and curved stiffener geometries.

Buckling analyses were performed using eigenvalue formulations in both NASTRAN and ABAQUS. The study first examined plates subjected to uniform and nonuniform temperature gradients, including through-thickness variations. These analyses were conducted for VAT laminates without stiffeners and subsequently extended to configurations with curvilinear stiffeners. All cases assumed simply supported boundary conditions.

An optimization framework was developed to maximize buckling performance by parameterizing VAT fiber paths using spline functions, along with the geometry of curvilinear stiffeners. This allowed for tailored redistribution of in-plane stresses.

Results showed that VAT laminates and curvilinear stiffeners can significantly influence stress distribution and alter buckling mode shapes, leading to improved buckling resistance.

This work highlights an advanced approach to tailoring composite structural performance through fiber steering and geometric optimization. The use of eigenvalue buckling is appropriate for computationally intensive optimization studies; however, temperature-dependent material behavior was not considered, which may affect the physical accuracy of the optimized designs at elevated temperatures.

A. Lakshmi Narayana et al. [2] investigated the thermal buckling behavior of composite plates with various cutout geometries.

The study employed the finite element method to analyze the effects of square and rectangular cutouts, as well as circular and elliptical cutouts at varying orientations. Additional parameters included plate aspect ratio, length-to-thickness ratio, and quasi-isotropic stacking sequences. ANSYS was used for all simulations, utilizing SHELL281 elements with eight nodes and six degrees of freedom per node. The laminates consisted of 16 plies with orientations spaced at 45° increments. Example stacking sequences included $[15/60/-30/-75]_{2s}$, $[30/75/-15/-60]_{2s}$, and $[45/90/0/-45]_{2s}$. Eigenvalue buckling analysis was used to evaluate critical buckling temperatures, and buckling mode shapes were not examined in detail.

Results showed that, for plates with square cutouts, the $[75/-60/30/-15]_{2s}$ stacking sequence exhibited the highest thermal buckling resistance across all cutout sizes. For plates with circular and elliptical cutouts, the critical buckling temperature decreased as the plate aspect ratio increased. A similar trend was observed for plates with square and rectangular cutouts.

This work demonstrates the significant influence of cutout geometry and stacking sequence on thermal buckling performance. However, temperature-dependent material properties were not considered in the analysis. As a result, some predicted buckling temperatures exceeded 500°C , raising questions about the physical realism of the results. Incorporating temperature-dependent material behavior may lead to more accurate predictions of buckling response under thermal loading.

A. R. Shaterzadeh et al. [3] investigated the thermal buckling behavior of rectangular composite plates with circular cutouts. Circular cutouts were considered due to their tendency to introduce nonuniform stress distributions, making analytical solutions difficult.

The study employed the finite element method using quadrilateral shell elements, with five degrees of freedom per node. ABAQUS software was used for all simulations. The analysis focused on four-layer antisymmetric square laminates with a $[45_G/-45_B]_2$ stacking sequence, where G denotes glass/epoxy plies and B denotes boron/epoxy plies. Each plate had dimensions of $100\text{ mm} \times 100\text{ mm}$ with a thickness of 0.25 mm and included a circular cutout of 10 mm diameter. Both two-edge and four-edge clamped boundary conditions were considered, and a mesh consisting of 432 elements was used. Temperature-dependent

material properties were not included in the analysis.

Results indicated that the critical buckling temperature increased with the number of layers for both symmetric and antisymmetric laminates. Additionally, the critical temperature generally increased with increasing hole size. Plates with two clamped edges were found to be more sensitive to stacking sequence than plates with all edges clamped. The location of the cutout also influenced behavior, with higher critical temperatures observed when the hole was positioned closer to clamped edges.

This study did not account for geometric nonlinearities or temperature-dependent material degradation, which may limit the accuracy of the predicted buckling response. Furthermore, the use of hybrid glass/epoxy and boron/epoxy laminates represents a relatively specialized configuration, which may limit the general applicability of the results. Nevertheless, the work provides useful insight into the influence of cutout geometry and boundary conditions on thermal buckling behavior.

Ranjan K. Behera et al. [4] investigated the free vibration and flexural behavior of clamped laminated composite plates (CLCP) under variable thermal environments. Clamped boundary conditions were considered due to the increased analytical complexity relative to simply supported plates, which are more commonly studied in the literature.

The study utilized the finite element method in ABAQUS, modeling the plates with shell elements. Three thermal loading cases were considered: a uniform temperature distribution, a linearly varying temperature across the plate, and a half-sine wave temperature gradient. The plates had dimensions of $600 \times 400 \times 5 \text{ mm}^3$ with a stacking sequence of $[0/90/0/90/0]$. Eight-noded quadratic shell elements were used, with six degrees of freedom per node. A 20×20 mesh (400 elements) was employed. Eigenvalue analysis was used to determine natural frequencies and mode shapes. Material properties were assumed to be constant.

Results included the natural frequencies of the plates under varying thermal conditions and mode numbers, as well as the distribution of von Mises thermal stresses along the plate edges.

The maximum temperature considered in this study was approximately 80°C , suggesting that temperature-dependent material effects were likely minimal. While the study provides insight into the influence of thermal gradients on vibration and stress behavior, it does not account for material degradation or nonlinear effects that may become significant at higher temperatures.

C. Sarath Babu et al. [5] investigated higher-order finite element models for the thermal buckling analysis of laminated composite and sandwich plates. Two formulations were developed, incorporating nine and eleven degrees of freedom per node, respectively.

The study introduced enhanced shell element formulations that include higher-order dis-

placement fields. The nine-degree-of-freedom model extends the classical five degrees of freedom $(u, v, w, \theta_x, \theta_y)$ by incorporating additional higher-order terms $(u^*, v^*, \theta_x^*, \theta_y^*)$, representing Taylor series expansions through the plate thickness. The eleven-degree-of-freedom model further includes θ_z and w^* to capture additional deformation effects. Eigenvalue buckling analyses were performed using these formulations.

A 6×6 mesh (36 elements) was used to evaluate both simply supported and clamped plate configurations. Results from the higher-order shell elements were compared with three-dimensional elasticity solutions to assess accuracy.

Results indicated that the nine-degree-of-freedom formulation provided significantly agreement with three-dimensional elasticity solutions, however they were not compared to classical shell models. The eleven-degree-of-freedom formulation showed only marginal improvement beyond the nine-degree-of-freedom model, suggesting diminishing returns with increased model complexity.

This work highlights the influence of element formulation on the accuracy of finite element predictions for composite laminates. However, the relatively coarse mesh used in the analysis, and the absence of a reported mesh convergence study, introduce some uncertainty regarding the numerical accuracy of the results. Also, the results were not compared to classical shell models, so the value of these models are unclear.

Saeedeh Qaderi et al. [6] investigated the use of Lagrange multiplier and penalty methods for the thermal buckling analysis of higher-order composite laminates.

Their work focused on graphene nanoplatelet-reinforced composites (GPLRC), in which graphene platelets are dispersed within an epoxy matrix. The analysis employed a general third-order shear deformation theory to capture higher-order effects. Several graphene distribution patterns were considered, corresponding to varying concentrations of graphene platelets through the thickness of the laminate.

A 20×20 mesh (400 elements) was used for the plate models. Eigenvalue buckling analysis was performed to determine critical buckling temperatures, with results presented in nondimensional form. Parametric studies were conducted to evaluate the influence of mesh density, thickness-to-length ratio, and thermal loading conditions. Material properties were assumed to be temperature independent.

Results demonstrated that the distribution of graphene platelets through the thickness significantly affects the critical buckling temperature. Functionally graded distributions generally improved buckling performance compared to uniform distributions.

This work highlights the behavior of composite materials with dispersed nanoscale reinforcement, which tend to exhibit more isotropic properties compared to traditional laminated composites. However, temperature-dependent material behavior and nonlinear effects were not considered. Additionally, the use of nondimensional temperature limits direct compari-

son with physically realistic temperature ranges.

Qingyuan Chen et al. [7] investigated the post-buckling behavior of composite plates subjected to combined in-plane compression and shear loading using the finite strip method. While post-buckling is commonly analyzed under pure compressive loading, the inclusion of shear loading introduces additional complexity and is less frequently studied.

The analysis was based on the first-order shear deformation theory (FSDT), in which each strip element includes five degrees of freedom corresponding to in-plane displacements, transverse displacement, and rotations, given by u , v , w , θ_x , and θ_y . Here, u and v represent the in-plane displacements, w is the transverse displacement, and θ_x and θ_y are the rotations about the x and y axes, respectively.

The study utilized the finite strip method, which reduces computational cost by discretizing the plate into longitudinal strips rather than full two-dimensional elements. Numerical examples were performed using a carbon/epoxy T800-3900-2 laminate, with a mesh consisting of 1600 equal-length strip elements. Parametric studies were conducted to evaluate the influence of fiber orientation on post-buckling response. Results were presented in terms of section forces versus central deflection, illustrating the nonlinear behavior of the plates beyond the initial buckling point.

This work highlights the applicability of the finite strip method as an efficient alternative to traditional finite element approaches for analyzing composite plate stability. Additionally, the study emphasizes post-buckling behavior under combined loading conditions, which is less commonly addressed in thermal buckling analyses that focus primarily on initial instability.

4 Theoretical Background

4.1 Micromechanical Models

To determine the effective thermomechanical properties of the composite, a combination of Rule of Mixtures, Chamis micromechanics relations, and Schapery's formulation for thermal expansion were employed. The longitudinal Young's modulus is approximated using the Rule of Mixtures.

$$E_1 = V_f E_f + V_m E_m \quad (1)$$

Where V_f is fiber volume fraction, E_f is fiber Young's modulus, V_m is matrix volume fraction, and E_m is matrix Young's modulus. The major Poisson's ratio is similarly estimated as.

$$\nu_{12} = V_f \nu_f + V_m \nu_m \quad (2)$$

where ν_f and ν_m are the fiber and matrix Poisson ratios, respectively.

The transverse modulus E_2 and in-plane shear modulus G_{12} are estimated using Chamis micromechanics relations,

$$E_2 = E_m \left(1 - \sqrt{V_f} + \frac{\sqrt{V_f}}{1 - \sqrt{V_f} \left(1 - \frac{E_m}{E_f} \right)} \right) \quad (3)$$

$$G_{12} = G_m \left(1 - \sqrt{V_f} + \frac{\sqrt{V_f}}{1 - \sqrt{V_f} \left(1 - \frac{G_m}{G_f} \right)} \right) \quad (4)$$

The coefficients of thermal expansion (CTE) are determined using Schapery's formulation.

$$\alpha_1 = \frac{E_m \alpha_m V_m + E_f \alpha_f V_f}{E_m V_m + E_f V_f} \quad (5)$$

$$\alpha_2 \approx \alpha_m + (\alpha_m - \alpha_1) \bar{\nu} - (\alpha_m - \alpha_f)(1 + \nu_f) V_f \quad (6)$$

Where α_1 is longitudinal CTE, α_2 is transverse CTE, α_m is matrix CTE, and α_f is fiber CTE.

4.2 Numerical Models

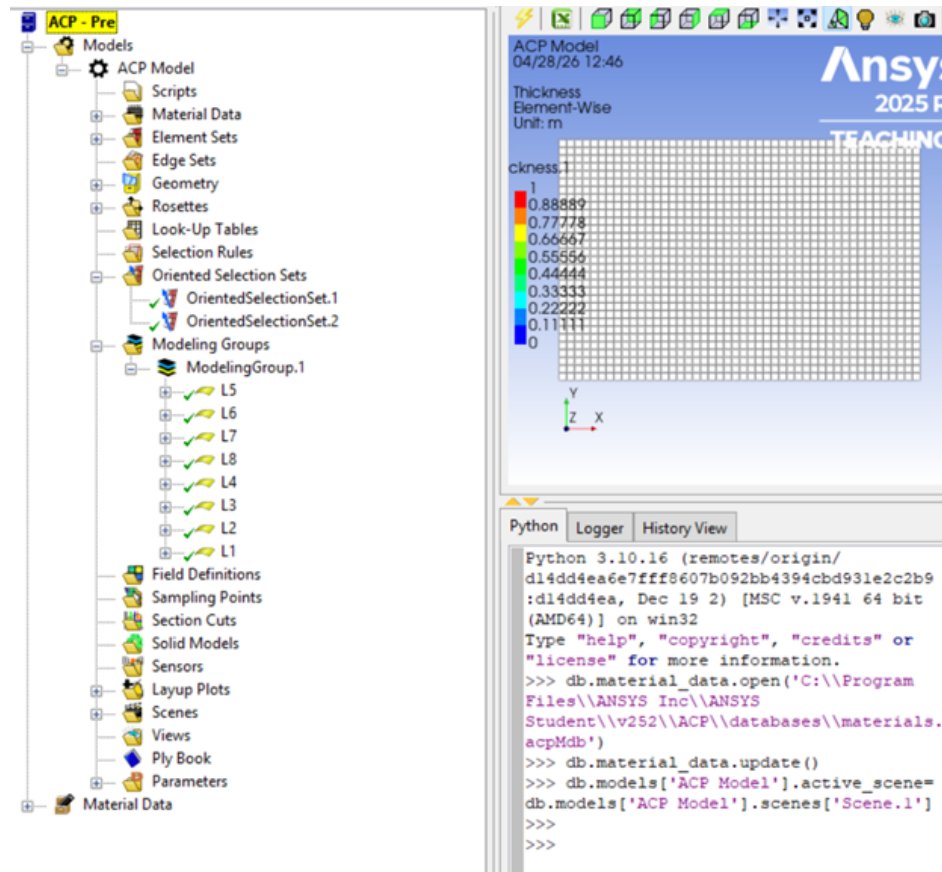


Figure 1: ANSYS ACP (Pre) Setup

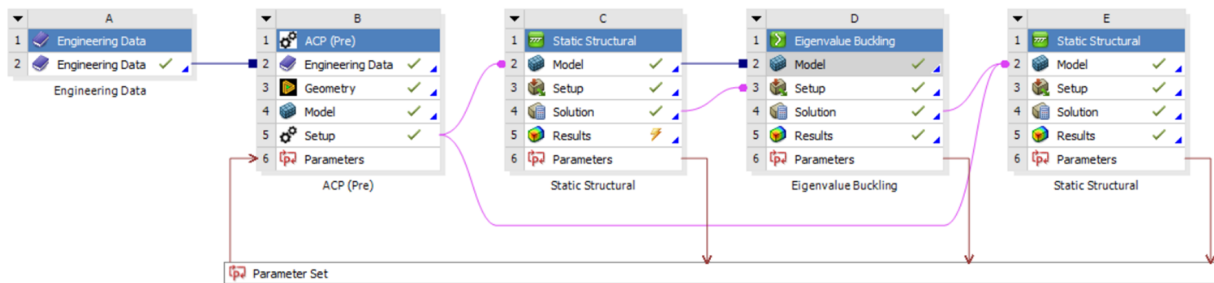


Figure 2: ANSYS Workbench Workflow

ACP (ANSYS Composite PrepPost) is a tool used to define composite material properties and layups. A screenshot of this platform can be seen in figure 1. In this work, ACP was used to setup and parameterize the stacking sequence of the laminate.

Eigenvalue buckling analysis determines the critical load by applying a preload from a static structural analysis, which is multiplied until the plate buckles 1 cm. This analysis returns a load multiplier corresponding to the instability point. Since eigenvalue buckling is a linear analysis, it does not directly account for temperature-dependent or nonlinear effects.

The buckling mode shape resulting from an Eigenvalue buckling analysis can be used as an initial geometric imperfection in a subsequent nonlinear analysis in static structural to capture more realistic structural behavior. The temperature is gradually increased until buckling is observed.

5 Material Properties

The fiber parameters are from the Hexcel Corporation on their Hextow IM7 carbon fiber product data sheet [8]. The matrix parameters are from the paper “Relaxation Model of the Relations between the Elastic Modulus and Thermal Expansivity of Thermosetting Polymers and FRPs” [9]. The Poisson ratio of each material was estimated as 0.35 for the matrix and 0.2 for the fiber. The Poisson ratio for the matrix was assumed to be constant with temperature. All parameters were assumed to be constant with temperature for the fiber, because the variance of the fiber parameters with temperature are minimal in relation to the variation in matrix parameters.

The material parameters of the matrix and fiber can be seen in Tables 1 and 2.

Table 1: Matrix (Epoxy) Properties

Temp (°C)	E (MPa)	ν_m	CTE (10^{-6} K^{-1})	G (MPa)
20	3300	0.35	21	1222
75	2700	0.35	26	1000
100	1700	0.35	38	630
130	164	0.35	57	60.7

Table 2: Fiber (Carbon) Properties

Temp (°C)	E (Pa)	V_f	CTE (ppm/°C)	G (Pa)
20	2.76×10^{11}	0.2	-0.64	1.15×10^{11}
75	2.76×10^{11}	0.2	-0.64	1.15×10^{11}
100	2.76×10^{11}	0.2	-0.64	1.15×10^{11}
130	2.76×10^{11}	0.2	-0.64	1.15×10^{11}

Using equations 1-6, the temperature dependent properties of the composite were derived from the parameters listed in Tables 1 and 2, and can be seen in Table 3 and Figure 3. In Ansys, these parameters are linearly interpolated to determine their values at a specific temperature.

Table 3: Composite Properties

Temp (°C)	E_1 (Pa)	E_2 (Pa)	ν_{12}	G_{12} (Pa)	α_1 (K ⁻¹)	α_2 (K ⁻¹)	ν_{23}	G_{23} (Pa)
20	1.67×10^{11}	1.16×10^{10}	0.26	4.33×10^9	-4.69×10^{-7}	1.10×10^{-5}	0.4	4.16×10^9
75	1.67×10^{11}	9.59×10^9	0.26	3.56×10^9	-4.67×10^{-7}	1.37×10^{-5}	0.4	3.42×10^9
100	1.66×10^{11}	6.10×10^9	0.26	2.27×10^9	-4.82×10^{-7}	2.02×10^{-5}	0.4	2.18×10^9
130	1.66×10^{11}	5.99×10^8	0.26	2.22×10^8	-6.17×10^{-7}	3.05×10^{-5}	0.4	2.14×10^8

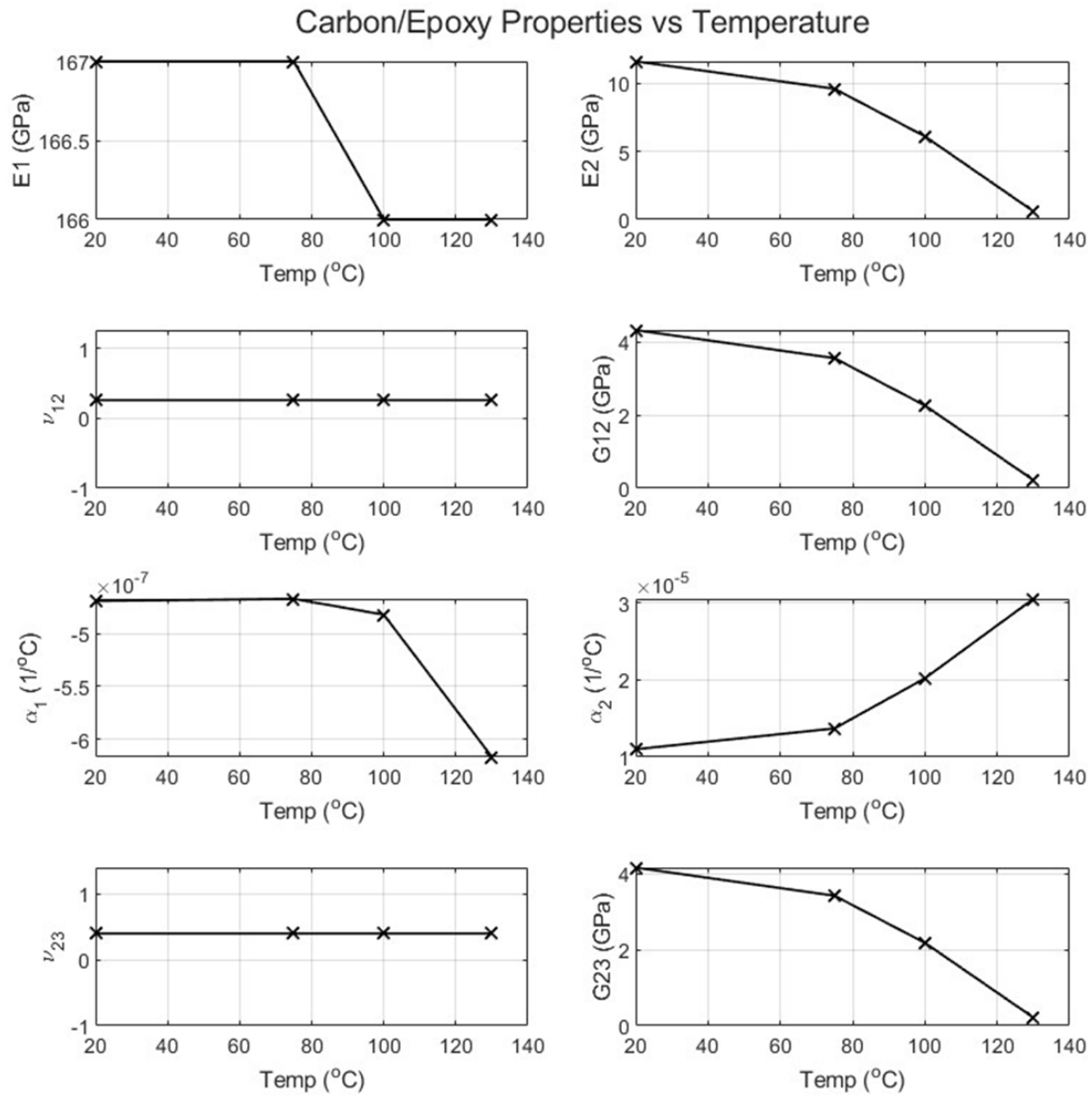


Figure 3: Carbon/Epoxy Composite Properties vs Temperature

6 Results and Discussion

6.1 Mesh Study

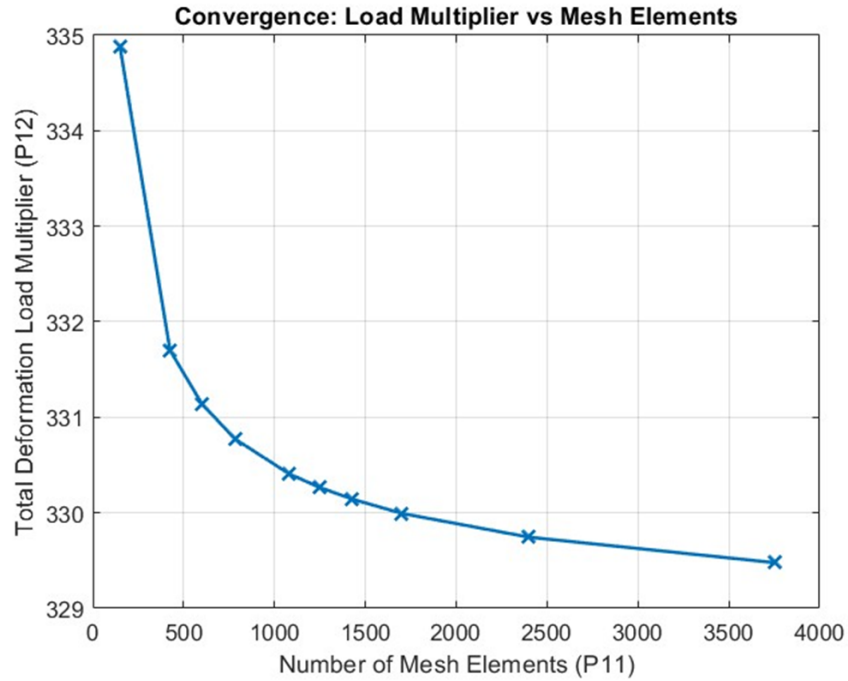


Figure 4: Mesh convergence study for the $[0/90/45/-45]_s$ laminate under eigenvalue buckling

A mesh convergence study was performed using a $[0/90/45/-45]_s$ laminate under eigenvalue buckling analysis. Simply Supported boundary conditions were imposed on a 400mm x 600mm composite plate with a thickness of 5mm. Each layer has a thickness of 0.75mm totaling 8 layers. The preload was generated in a static structural analysis with a uniform 1 °c increase over ambient temperature. This corresponds to a 23°C temperature condition on the plate, with an ambient temperature of 22°C.

The results of this mesh study can be seen in figure 4. Based on the results, a mesh density of approximately 1400 elements was determined to provide adequate refinement. This level of refinement is greater than that used in some reference studies (approximately 400 elements), providing increased confidence in the numerical accuracy of the results.

6.2 Eigenvalue Buckling

Eigenvalue Buckling analysis was performed on a matrix of 12 stacking sequences and 3 plate areas. The same setup and boundary conditions were imposed as with the mesh study described earlier. The stacking sequences were all relevant combinations of 0, 90, and ± 45 plies with symmetry. Since the plate is rectangular, 45 and -45 are interchangeable, so only the plates with 45 before -45 were studied, which results in a total of 12 quasi-isotropic stacking sequences.

Three plate sizes were studied as well, with 600x400mm, 900x600mm, and 1200x800mm, with corresponding areas of 0.24 m², 0.54 m², and 0.96 m². These dimensions all correspond to an aspect ratio of 1.5. The ratio of plate area to thickness changed depending on plate area, as the thickness of the plate remained at 5mm. The resulting area to thickness ratios were 2.5E-05, 1.11E-05, and 6.25E-06 m⁻¹.

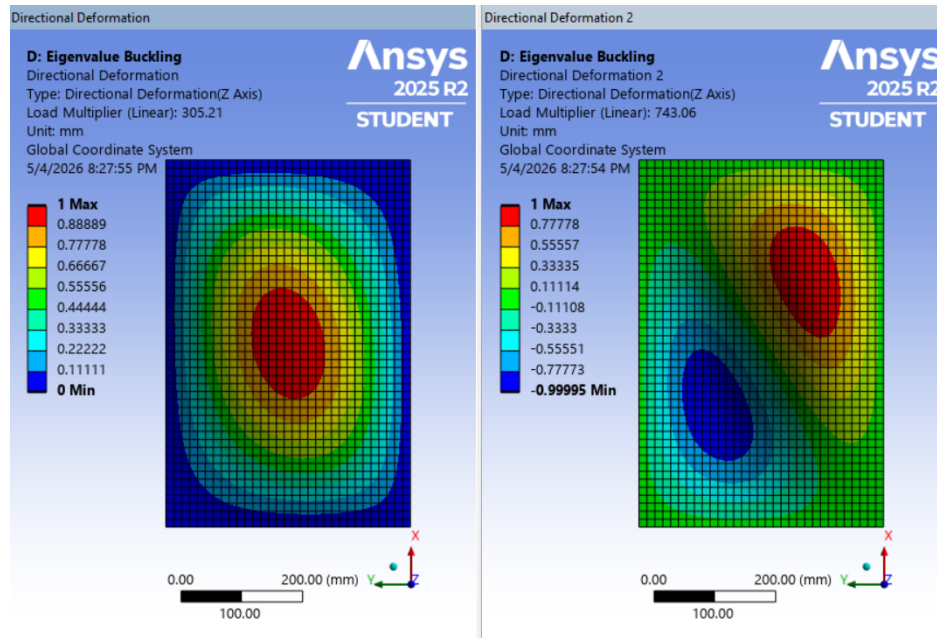


Figure 5: Mode 1 and 2 Buckling Mode Shapes for $[0/45/90/-45]_s$, and $A = 0.24 \text{ mm}^2$

The first and second buckling mode shapes for $[0/45/90/-45]_s$, and $A = 0.24 \text{ mm}^2$ can be seen in figure 5. Some asymmetry can be seen in mode 1, but it is a mostly symmetric buckling shape. All stacking sequences and area to thickness ratios had similar mode 1 shapes, with their own slight asymmetries. As this study primarily analyzes the minimum buckling temperature of the plates, not every mode shape was presented in this paper. The

second buckling mode shape can also be seen in figure 5. Each stacking sequence saw a similar mode 2 buckling shape with varying zero deflection line angles. Mode 2 was ignored for further results as it buckled at a higher temperature than mode 1.

Plate Area (m ²)	Area to Thickness Ratio (t/A)	[0, 90, 45, -45] _s	[0, 45, 90, -45] _s	[0, 45, -45, 90] _s	[90, 0, 45, -45] _s	[90, 45, 0, -45] _s	[90, 45, -45, 0] _s	[45, -45, 90, 0] _s	[45, -45, 0, 90] _s	[45, 90, 0, -45] _s	[45, 90, 0, -45] _s	[45, 0, -45, 90] _s	[45, 0, 90, -45] _s
0.24	2.50E-05	352.1	327.2	335.6	461.9	515.3	557.0	529.3	491.7	512.6	458.4	394.5	376.2
0.54	1.11E-05	169.4	158.7	162.8	218.4	242.8	261.8	251.5	234.7	242.9	218.2	190.0	181.3
0.96	6.25E-06	105.0	99.2	101.6	132.7	146.6	157.4	152.2	142.8	147.0	133.0	117.2	112.2

Most Effective: [90, 45, -45, 0]_s
Least Effective: [0, 45, 90, -45]_s
Median Effective: [45, 90, 0, -45]_s

Table 4: Eigenvalue Buckling Results

Table 4 shows the critical buckling temperature for all stacking sequences and area to thickness ratios. The most effective stacking sequence was [90, 45, -45, 0]_s, the least effective was [0, 45, 90, -45]_s, and the median effective sequence was [45, 90, 0, -45]_s, with all area to thickness ratios having the same results. As the area to thickness ratio decreased, the temperature required to buckle the plates decreased.

Stacking sequences with an outer fiber angle of 0 degrees performed the worst. This corresponds with longer fibers stretching along the longer edge of the rectangle. This means the direction with the strongest resistance to bending is along the y axis as seen in figure 5. The forces of buckling must then have the greatest bending moment in the x axis, which is why the strongest stacking sequences had 90 degree fiber on the outer surfaces.

6.3 Nonlinear Buckling

Nonlinear buckling analyses were performed in ANSYS. The boundary conditions of a static structural analysis had simply supported edges to a rectangular plate, a ramped temperature from 23-200°C, and an initial imperfection of 0.001 times mode 1 from the eigenvalue buckling results of the same plate. 150 sub steps were performed to determine the buckling temperature, which was indicated by the temperature at which the maximum deformation of

the plate reaches 1mm. The buckling shape of each analysis matched the eigenvalue buckling shape, one of which can be seen in figure 6.

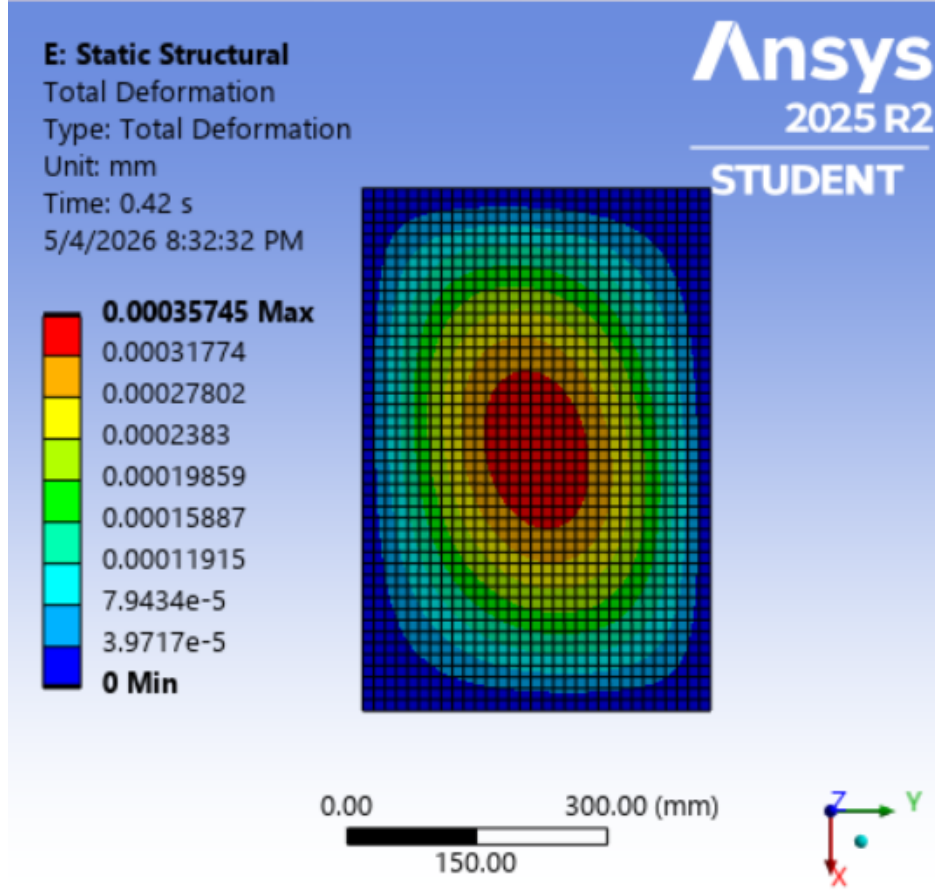


Figure 6: Nonlinear Buckling Shape for $[0/45/90/-45]_s$, and $A = 0.24 \text{ mm}^2$ at maximum deformation

Three nonlinear buckling analyses were performed on plates deemed as significant from the eigenvalue buckling results. The plates selected for these analysis were $[0, 45, 90, -45]_s$, with $t/A = 6.25\text{E-}6 \text{ m}^{-1}$ which was the weakest plate in the 36 point matrix seen in table 4, and had a buckling temperature of 99.2°C . The next plate studied was $[45, 90, 0, -45]_s$ with $t/A = 1.11\text{E-}5 \text{ m}^{-1}$, which was chosen because of its average results at a buckling temperature of 218.2°C . $[45, 90, 0, -45]_s$ with $t/A = 2.5\text{E-}5 \text{ m}^{-1}$ was the final plate studied because of its high eigenvalue buckling temperature of 458.4°C .

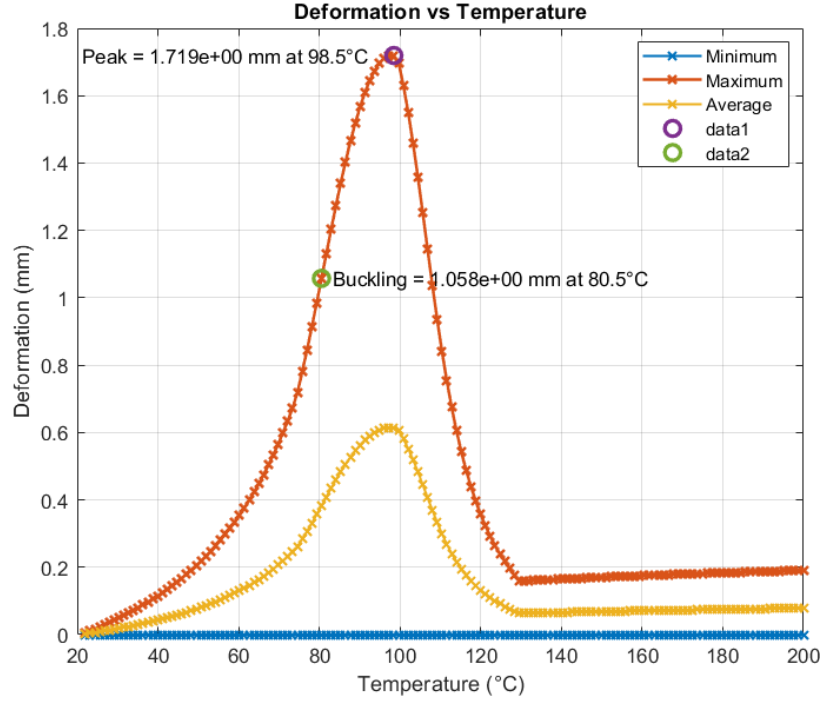


Figure 7: Nonlinear Buckling Results of $[0, 45, 90, -45]_s$, and $t/A = 6.25E-6 \text{ m}^{-1}$

Figure 7 shows the nonlinear buckling results of $[0, 45, 90, -45]_s$, and $t/A = 6.25E-6$. The plate reached a deformation of 1mm at a temperature of 80.5°C, which is the buckling temperature. This means that with nonlinear effects taken in account, the buckling temperature of this plate decreased by 18.7°C from 99.2°C to 80.5°C. These results should be considered carefully though, because as the temperature increased, the deformation of the plate decreased again back below 0.2mm. This suggests that the shifting material properties of the composite allows for the plate to flex back to flat. As the temperature increases past 200°C the plate should fail by some other means such as melting.

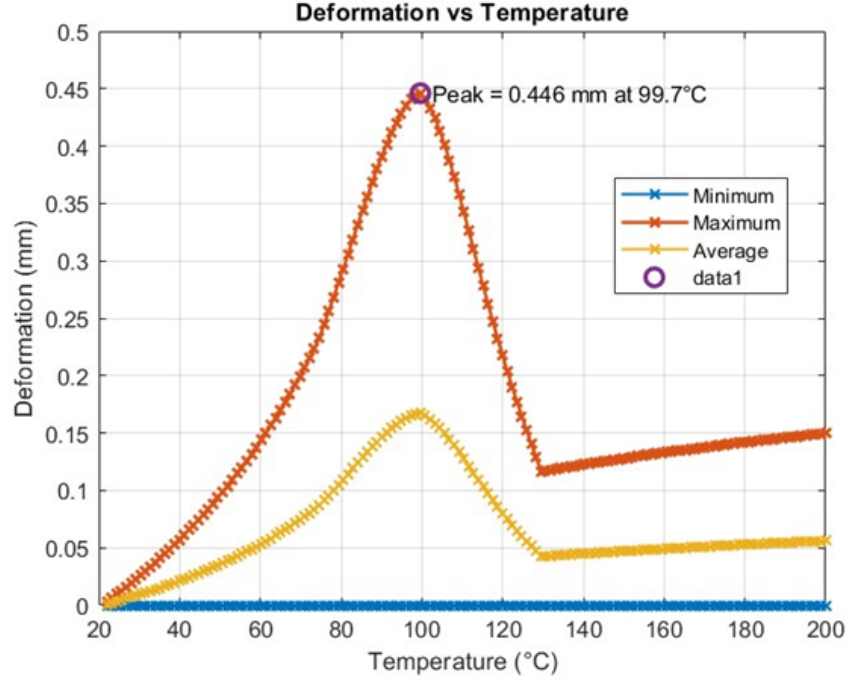


Figure 8: Nonlinear Buckling Results of $[45, 90, 0, -45]_s$, and $t/A = 1.11\text{E-}5 \text{ m}^{-1}$

Figure 8 shows the Nonlinear Buckling Results of $[45, 90, 0, -45]_s$, with $t/A = 1.11\text{E-}5 \text{ m}^{-1}$. This plate failed to reach a deformation of 1mm, so the buckling temperature was taken as the temperature of maximum deformation, which was 99.7°C. This temperature is far lower than the eigenvalue result of 218.2°C, which shows that temperature-dependent effects of the material can significantly influence the buckling temperature. The buckling temperature was also far closer to the plate with $t/A = 6.25\text{E-}6 \text{ m}^{-1}$. This suggests that the softening of the matrix is more important for predicting buckling than the area to thickness ratio of the plate.

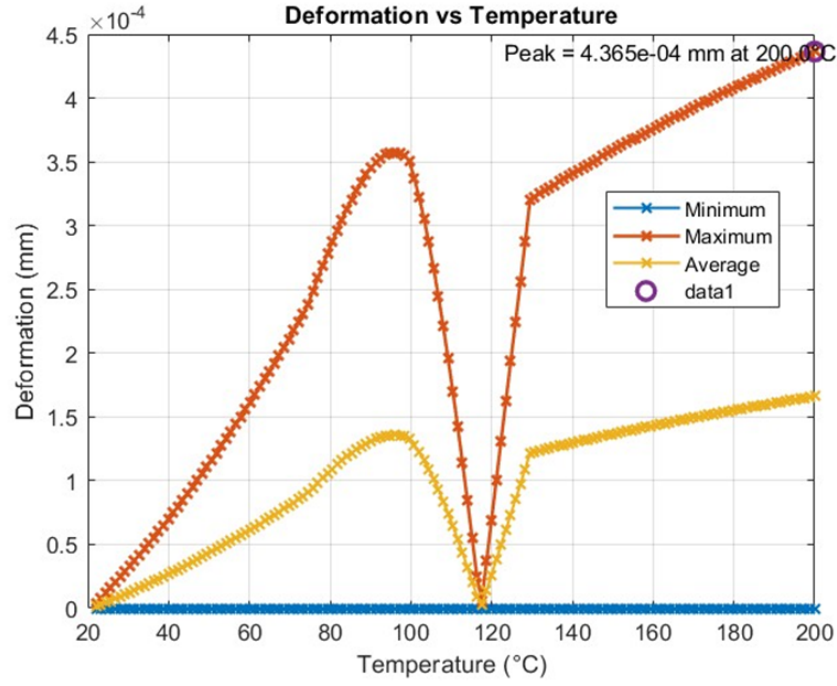


Figure 9: Nonlinear Buckling Results of [45, 90, 0, -45]s with $t/A = 2.5E-5 \text{ m}^{-1}$

The last nonlinear buckling result can be seen in figure 9. The [45, 90, 0, -45]s with $t/A = 2.5E-5 \text{ m}^{-1}$ plate had an eigenvalue buckling temperature of 458.4°C. This plate failed to buckle during this analysis, and had a maximum deformation of 3.465e-4 mm at a temperature of 200°C. The plate did not buckle at higher temperatures either. These results suggest that for plates with large thickness to area ratios, thermal buckling is not an important failure mode to consider.

7 Conclusions

The primary conclusions of this study are summarized as follows:

- Eigenvalue buckling analysis frequently predicted critical temperatures significantly higher than those achievable by the material, indicating that such results may not be physically realistic.
- A substantial difference was observed between eigenvalue and nonlinear buckling predictions. The eigenvalue approach, being linear, does not account for material and geometric nonlinearities, particularly temperature-dependent property degradation.
- Based on nonlinear buckling results, certain composite configurations did not exhibit buckling within the investigated temperature range. This suggests that failure may occur through other mechanisms, such as material degradation or strength limits, prior to buckling.
- Experimental validation is recommended to determine which analysis approach provides the most accurate representation of physical behavior.
- For simply supported boundary conditions, bending effects were found to be most significant along the longer dimension of the plate. As a result, orienting the outer fibers transverse to the longer edge of the plate may improve resistance to thermally induced buckling.

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